

INFORMATION AGGREGATION AND SOCIAL NETWORKS: RESPONSIVENESS AND OVERTURNING

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INTRODUCTION

MOTIVATION

- ▶ When making decisions under uncertainty, individuals often **learn from others**.
 - ▶ For example, people rely on others' choices when adopting a new technology.
- ▶ What individuals learn from others depends on **whom they can observe**.
- ▶ Digitization expands access to others' behavior and reshapes **observation networks**.
- ▶ Experimental evidence further illustrates the practical relevance of social observation.
 - ▶ Social influence can weaken the wisdom of crowds (Lorenz *et al.*, 2011; Muchnik *et al.*, 2013).
 - ▶ Yet, social observation can improve collective accuracy (Becker *et al.*, 2017; Cai *et al.*, 2009).
- ▶ **How do network structures affect information aggregation?**

MODEL

- ▶ There are finite $N \geq 3$ **agents**, where agent i chooses $a_i \in A = \{0, 1\}$ in period i .
- ▶ There are binary **states** $\theta \in \Theta = \{L, H\}$ with **prior** $\mu_i \in (0, 1)$ for each i .
- ▶ Utility: $u(0, L) = u(1, H) = 1$, and the payoff is zero otherwise.
- ▶ In each period i , agent i observes the **history** $(a_j)_{j \in \mathcal{N}_i}$ and her **private signal** s_i .
 - ▶ **Network** $\mathcal{N} = (\mathcal{N}_i)_{i=1}^N$: $\mathcal{N}_i \subseteq \{1, \dots, i-1\}$ specifies agent i 's neighborhood.
 - ▶ Private signal s_i is (independently) drawn from the **information structure** $\Pi_i = (S_i, \pi_i)$.
 - ▶ (μ, Π) denotes the **informational environment**.

OVERVIEW

- ▶ Thm 1: Which networks can be justified by some informational environment?
 - ▶ Network efficiency depends critically on the informational environment.
 - ▶ Thus, **connectivity has an ambiguous social value** despite its positive private value.
- ▶ Thms 2 & 3: What forces and mechanisms determine the network efficiency?
 - ▶ **Responsiveness**: sparse links preserve responsiveness of actions to private information.
 - ▶ **Overturning**: dense links reveal strong signals that reverse public beliefs.
- ▶ Prop 3 & numerics: What features of information structure determine this trade-off?
 - ▶ The trade-off depends on **the relative probability of highly accurate information**.

LITERATURE

- ▶ Social learning/observational learning: (see [Bikhchandani *et al.* \(2024\)](#) for a survey)
 - ▶ Pioneering work: [Banerjee \(1992\)](#), [Bikhchandani *et al.* \(1992\)](#), and [Smith and Sørensen \(2000\)](#).
- ▶ Social learning in social networks: (see [Golub and Sadler \(2016\)](#) for a survey)
 - ▶ [Gale and Kariv \(2003\)](#), [Çelen and Kariv \(2004\)](#), [Acemoglu *et al.* \(2011\)](#), [Lobel and Sadler \(2015\)](#), [Mossel *et al.* \(2015\)](#), [Arieli and Mueller-Frank \(2019, 2021\)](#), [Dasaratha *et al.* \(2023\)](#), [Kartik *et al.* \(2024\)](#).
 - ▶ [SgROI \(2002\)](#): "Guinea pigs" can increase social welfare.
 - ▶ [Dasaratha and He \(2019\)](#): Comparisons of information loss due to confounding.
- ▶ Non-Bayesian frameworks:
 - ▶ [Ellison and Fudenberg \(1993, 1995\)](#), [Bala and Goyal \(1998, 2001\)](#), [DeMarzo *et al.* \(2003\)](#), [Golub and Jackson \(2010, 2012\)](#), [Jadbabaie *et al.* \(2012\)](#), and [Molavi *et al.* \(2018\)](#)

MAIN ANALYSIS

EVALUATION CRITERION

- ▶ We evaluate network $\mathcal{N}$ by the **focal agent's payoff** $V_N(\mathcal{N}, \mu, \Pi)$ (or social welfare).
 - ▶ In a finite society, agent N may correspond to the ultimate decision maker.
 - ▶ In an infinite society, N represents a finite learning horizon.
- ▶ Much of the literature evaluates networks by whether asymptotic learning occurs.
 - ▶ Whether asymptotic learning occurs classifies networks into only two categories.
 - ▶ The challenge is to compare exact expected payoffs, rather than asymptotic welfare.
- ▶ When signals (not actions) are observed, the predecessor network does not matter.
- ▶ When actions (not signals) are observed, the predecessor network affects inference.

RATIONALIZABLE NETWORKS

- ▶ In practice, informational environments can vary across economic settings.
- ▶ **Which networks can be justified from the perspective of information aggregation?**
- ▶ $\hat{\mathcal{N}}$ is **rationalizable** if, for some (μ, Π) , $V_N(\hat{\mathcal{N}}, \mu, \Pi) > V_N(\mathcal{N}, \mu, \Pi)$ for every $\mathcal{N} \neq \hat{\mathcal{N}}$.

Theorem 1

$\hat{\mathcal{N}}$ is rationalizable if and only if $\hat{\mathcal{N}}_N = \{1, \dots, N-1\}$.

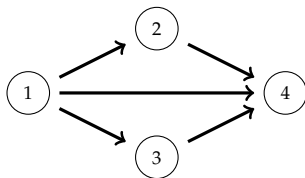
- ▶ **Thus, network efficiency depends critically on the informational environment.**
- ▶ This offers a rationale for the many different forms of networks in practice.

COROLLARY 1: SOCIAL VALUE OF CONNECTIVITY

- ▶ Fix a link $j \rightarrow i$ among predecessors, where $j < i < N$.
- ▶ There is an informational environment in which **adding this link increases** V_N .
- ▶ There is an informational environment in which **removing this link increases** V_N .
 - ▶ The link is weakly valuable to agent i , who gains an observation.
 - ▶ But it changes how agent i combines private information and social observation.
 - ▶ Her action may therefore become more or less informative to later agents.
- ▶ Thus, **connectivity has an ambiguous social value** despite its positive private value.

PROOF SKETCH OF THEOREM 1

- Necessity is immediate, and the substantive challenge is the sufficiency direction.
 - One environment must **uniformly** favor the target network over **all** alternatives.



Target shield network

1. Choose **priors** for indifference at the target observation.
2. Give agents 2 and 3 two weak **probe signals**.
 - Only the target neighborhood & history separate them.
3. Give agent 4 **test signals** that make separation valuable.
4. Use further smaller perturbations at each step.

SOCIALLY RATIONALIZABLE NETWORKS

- ▶ Fix weights $\lambda = (\lambda_1, \dots, \lambda_N) \in \mathbb{R}_{++}^N$: define $W_\lambda(\mathcal{N}, \mu, \Pi) = \sum_{i=1}^N \lambda_i V_i(\mathcal{N}, \mu, \Pi)$.
 - ▶ For example, $\lambda_i = \delta^{i-1}$ for a discounted sum.
- ▶ $\hat{\mathcal{N}}$ is **socially rationalizable** if $\exists(\mu, \Pi), W_\lambda(\hat{\mathcal{N}}, \mu, \Pi) > W_\lambda(\mathcal{N}, \mu, \Pi), \forall \mathcal{N} \neq \hat{\mathcal{N}}$.

Proposition 1

$\hat{\mathcal{N}}$ is socially rationalizable if and only if $\hat{\mathcal{N}}_N = \{1, \dots, N-1\}$.

- ▶ The characterization is unchanged under positive welfare weights.

TWO BENCHMARK NETWORKS

- ▶ **What forces and mechanisms determine the network efficiency?**
 - ▶ A general construction behind Theorem 1 cannot isolate underlying mechanisms.
- ▶ We revisit two networks at opposite ends of connectivity.
 - ▶ **Star network:** $\mathcal{N}_i^S = \emptyset$ for $i < N$ and $\mathcal{N}_N^S = \{1, \dots, N - 1\}$.
 - ▶ **Complete network:** $\mathcal{N}_i^C = \{1, \dots, i - 1\}$ for every i .
- ▶ From now on, agents share a prior of $1/2$ and a common information structure Π .

STAR NETWORK

Theorem 2

There exists Π^B such that $V_N(\mathcal{N}^S, \Pi^B) > V_N(\mathcal{N}, \Pi^B)$ for every $\mathcal{N} \neq \mathcal{N}^S$.

- ▶ Under binary signals, an isolated agent's action perfectly reveals her signal.
 - ▶ Agent N infers every predecessor's private signal under $\mathcal{N}^S$.
- ▶ Links among predecessors can immediately generate information cascades.
 - ▶ A connected agent may follow history and ignore her private signal.
 - ▶ Her action then becomes less responsive and less informative.
- ▶ The star network preserves **responsiveness** of predecessors' actions.

COMPLETE NETWORK

Theorem 3

There exists Π^* such that $V_N(\mathcal{N}^C, \Pi^*) > V_N(\mathcal{N}, \Pi^*)$ for every $\mathcal{N} \neq \mathcal{N}^C$.

- ▶ Four signals: l, h (ordinary) & l^*, h^* (strong), where l, l^* favor L and h, h^* favor H .
- ▶ Under $\mathcal{N}^C$, successive action reversals reveal stronger signals.
 - ▶ A history of action 1 is likely generated by the ordinary signal h .
 - ▶ A subsequent action 0 indicates the more informative negative signal l .
 - ▶ A further action 1 indicates still stronger good news, namely h^* .
 - ▶ A final action 0 similarly indicates the strongest bad news l^* .
- ▶ Complete observation makes each **overturning** publicly interpretable.

TWO EFFECTS

- ▶ The **responsiveness effect** favors sparse networks.
 - ▶ Fewer observations weaken informational interdependence among actions.
 - ▶ Keeping actions responsive to private signals mitigates cascades.
- ▶ The **overturning effect** favors dense networks.
 - ▶ Rich public histories identify signals that overturn previous actions.
 - ▶ Overturning corrects histories that otherwise induce incorrect inference.
- ▶ **This trade-off determines whether sparse or dense networks perform better.**

RESPONSIVENESS AND OVERTURNING

FRAMEWORK

- ▶ Which features of information structures determine which effect dominates?
- ▶ M agents (**guinea pigs**) act independently; later agents observe all preceding actions.
 - ▶ For $M \in \{1, \dots, N-1\}$, $\mathcal{N}_i^M = \emptyset$ for $i \leq M$ and $\mathcal{N}_i^M = \{1, \dots, i-1\}$ for $i > M$.
 - ▶ As M increases, $\mathcal{N}^M$ becomes sparser, from complete ($M=1$) to star ($M=N-1$).
- ▶ Let $\Pi^{\varepsilon,p} = (\{l^*, l, h, h^*\}, \pi)$ for $p \in (1/2, 1)$ and $\varepsilon \in (0, 1)$:
$$\begin{aligned}\pi(l \mid L) &= \pi(h \mid H) = (1 - \varepsilon)p, & \pi(h \mid L) &= \pi(l \mid H) = (1 - \varepsilon)(1 - p), \\ \pi(l^* \mid L) &= \pi(h^* \mid H) = \varepsilon, & \pi(h^* \mid L) &= \pi(l^* \mid H) = 0,\end{aligned}$$
 - ▶ Ordinary signals l, h have accuracy p ; conclusive signals l^*, h^* occur with probability ε .
- ▶ We derive the number of guinea pigs M^* maximizing agent N 's payoff $V_N(\mathcal{N}^M, \Pi^{\varepsilon,p})$.

THE OPTIMAL NUMBER OF GUINEA PIGS

Proposition 3

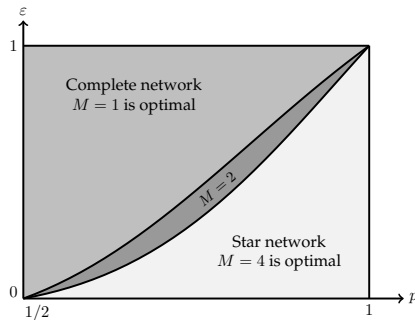
Suppose that $N \geq 3$ is odd. Then, there are unique cutoffs $1 > \varepsilon_0(p) > \varepsilon_1(p) > \dots > 0$.
For $1 \leq k < (N - 1)/2$, the optimum is $M^* = 2k$ whenever $\varepsilon_k(p) < \varepsilon < \varepsilon_{k-1}(p)$.

- ▶ Each $\varepsilon_k(p)$ is strictly increasing in p .
- ▶ Cutoff for the complete network:

$$\varepsilon_0(p) = \frac{p(2p-1)}{p+2(1-p)^2}.$$

- ▶ Limiting cutoff for the star network:

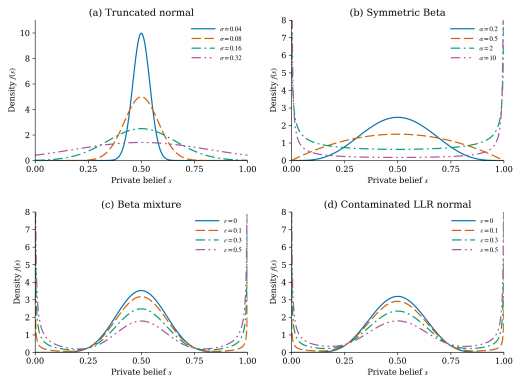
$$\varepsilon_k(p) \xrightarrow{k \rightarrow \infty} \frac{(2p-1)^2}{1+4(1-p)^2} \in (0, 1).$$



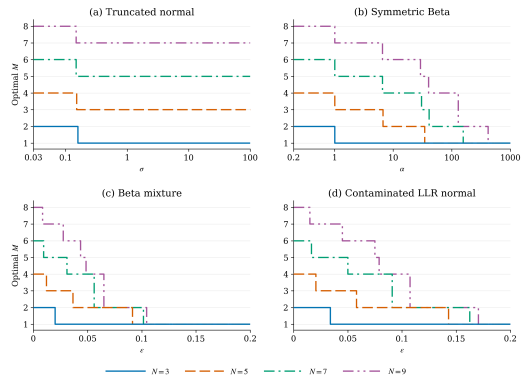
Optimal number of guinea pigs for $N = 5$

NUMERICAL EXERCISES

Private-belief distributions



Optimal number of guinea pigs



RESPONSIVENESS AND OVERTURNING

- ▶ The key determinant is the **relative tail distribution** of private signals.
- ▶ **Moderately informative signals** favor preserving many guinea pigs.
 - ▶ Preserving the responsiveness of each action is then particularly valuable.
- ▶ **Frequent extreme signals** make overturning more valuable.
 - ▶ Extreme signals occur often enough to be identified through overturning histories.
- ▶ **In practice, the key is how often people receive highly accurate information.**

CONCLUSION

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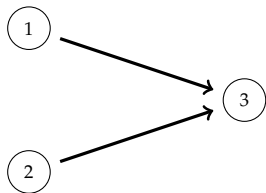
- ▶ We characterize networks that can be justified under some information environment.
 - ▶ The same characterization holds for positively weighted social welfare.
 - ▶ **The efficiency of networks depends critically on the informational environment.**
 - ▶ This offers a rationale for the many different forms of networks in practice.
 - ▶ A predecessor link has an **ambiguous social effect** despite its positive private value.
- ▶ We identify a **trade-off** shaping the network efficiency of information aggregation.
 - ▶ Sparse networks preserve **responsiveness** to private signals.
 - ▶ Dense networks identify informative signals through the **overturning effect**.
 - ▶ The **relative tail distribution** determines the balance between the two effects.

THE END.

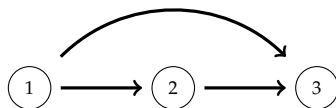
APPENDIX

ILLUSTRATIVE EXAMPLES: THREE-AGENT NETWORKS

Star network



Complete network



- ▶ **Star network:** agents 1 and 2 act only on their private signals.
- ▶ **Complete network:** agent 2's action also responds to agent 1's action.
- ▶ If signals (not actions) were observable, both networks would inform agent 3 equally.

EXAMPLE 1: RESPONSIVENESS EFFECT

Binary information structure

$$\pi(h | H) = \pi(l | L) = 9/10 \text{ and } \pi(l | H) = \pi(h | L) = 1/10.$$

- ▶ In the star network, a_1 and a_2 perfectly reveal s_1 and s_2 .
- ▶ In the complete network, agent 2 may follow agent 1 and ignore s_2 .
 - ▶ Hence, agent 2's distinct private signals can lead to the same action.
 - ▶ At $(s_1, s_2, s_3) = (l, h, h)$, $a_3 = 1$ in the star network but $a_3 = 0$ in the complete network.
- ▶ The star network gives agent 3 a strictly higher expected payoff.
- ▶ Intuition: Sparse networks preserve **responsiveness** of actions to private signals.

EXAMPLE 2: OVERTURNING EFFECT

— Ordinary and conclusive positive signals —

$S = \{l, h, h^*\}$, where h^* perfectly reveals H .

- ▶ In the star network, action 1 pools the signals h and h^* .
- ▶ In the complete network, agent 2 conditions on agent 1's action.
 - ▶ After $a_1 = 0$, only h^* induces $a_2 = 1 \implies (a_1, a_2) = (0, 1)$ reveals the conclusive signal.
- ▶ The complete network gives agent 3 a strictly higher expected payoff.
- ▶ Intuition: Dense networks identify strong signals from **overturning** histories.

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